ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE J

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. Comparators, voltage dividers with sensors and regulators, encoders, motor power and torque, hybrid control, hydraulic cylinder force and speed and accumulator safety, a PLC ladder program with a timer and a counter for an electro-hydraulic press, and maintenance strategies, oil sampling and replacing seals.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL calculations. Draw all circuits and symbols neatly, using a ruler and standard symbols.
5. Write neatly and legibly.

QUESTION 1: ELECTRONICS

1.1 Explain how an operational amplifier works as a comparator. (4)

1.2 A thermistor and a 10 kΩ resistor form a voltage divider across 5 V. At a certain temperature the thermistor's resistance is 15 kΩ. Calculate the voltage across the 10 kΩ resistor. (3)

1.3 Explain the purpose of a 7805 voltage regulator, and why it needs capacitors at its input and output. (4)

1.4 Explain how a transistor amplifies a signal in a common-emitter amplifier. (4)

1.5 Give FIVE components that are found in most electronic control circuits. (5)

[20]

QUESTION 2: DRIVES AND CONTROL

2.1 Explain the difference between an incremental encoder and an absolute encoder. (4)

2.2 A motor delivers a torque of 8 N·m at 1 500 rpm. Calculate its output power. (3)

2.3 Give THREE factors to consider when choosing a motor for a robot joint. (3)

2.4 Explain what hybrid control is in a mechatronic system, and give ONE example. (4)

2.5 Explain the difference between a program and programming. (4)

2.6 Give TWO advantages of a servo motor over a stepper motor. (2)

[20]

QUESTION 3: HYDRAULICS

3.1 A hydraulic cylinder has a piston area of 0,005 m² and works at 8 MPa. Calculate the force it produces. (3)

3.2 The cylinder in QUESTION 3.1 is supplied with oil at 0,0005 m³/s. Calculate its extending speed. (3)

3.3 Explain why an accumulator must be discharged before work is done on a hydraulic system. (3)

3.4 Explain the difference between a pressure-compensated flow control valve and a simple throttle valve. (4)

3.5 Give FOUR safety rules for working with hydraulic systems. (4)

3.6 Name THREE functions of a hydraulic reservoir (tank). (3)

[20]

QUESTION 4: PLC CONTROL OF AN ELECTRO-HYDRAULIC PRESS

4.1 Explain how an up-counter in a PLC works. (3)

4.2 A press cylinder is controlled by a 4/2-way double-solenoid valve: Q0.0 (Y1) extends it and Q0.1 (Y2) retracts it. Inputs: START I0.0 (NO), STOP I0.1 (NC), limit switch S2 at full extension I0.2. Draw a PLC ladder program so that: START latches a run memory M0.0; while running, the cylinder extends; at full extension timer T1 holds the press for 2 s; then the cylinder retracts; counter C1 counts each press cycle and stops the machine after 5 cycles. (13)

4.3 Give FOUR advantages of controlling the press with a PLC instead of relays. (4)

[20]

QUESTION 5: MAINTENANCE

5.1 Explain the following maintenance strategies: (8)

5.1.1 Reactive (breakdown) maintenance (2)

5.1.2 Preventive maintenance (2)

5.1.3 Predictive (condition-based) maintenance (2)

5.1.4 Total productive maintenance (TPM) (2)

5.2 Explain why oil samples are taken from hydraulic systems and what can be learned from them. (4)

5.3 Describe, step by step, how to replace the seals of a hydraulic cylinder. (6)

5.4 Give TWO items of information that must be written on a fault report. (2)

[20]

TOTAL: 100

END OF QUESTION PAPER